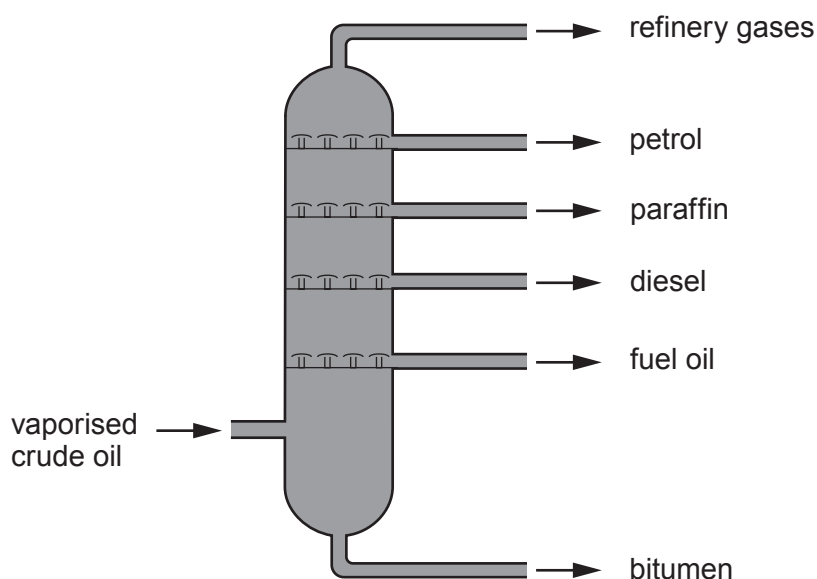


3. Crude oil is a mixture of hydrocarbon compounds. It is separated into different fractions inside a fractionating column. Each fraction has a different boiling point range.



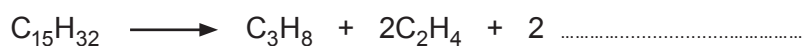
- (a) Explain how the length of the hydrocarbon chains within each fraction determines where each fraction collects in the column. [2]

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- (b) Many of the hydrocarbons that are collected from crude oil go through a second process called cracking. The equation shows the cracking of the hydrocarbon with the formula $C_{15}H_{32}$.



- (i) **Complete the equation** for this reaction by adding the **formula** of **one** other product. [1]

- (ii) Give the reaction conditions used in the process. [1]

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- (iii) Give **one** reason why it is important for oil companies to carry out cracking. [1]

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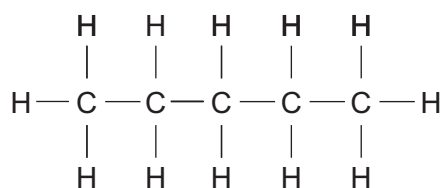


(c) Many hydrocarbons display isomerism.

(i) Give the meaning of the term *isomer*.

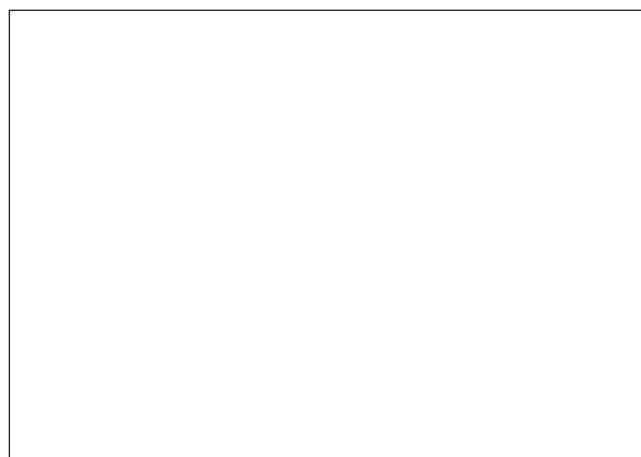
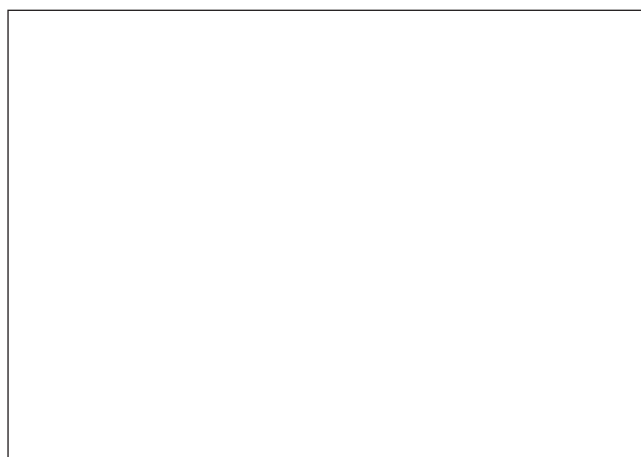
[1]

(ii) C_5H_{12} has three isomers. The diagram shows one of these isomers.



Draw the **two** other isomers of C_5H_{12} .

[2]



- (d) 8.4 g of a hydrocarbon was found to contain 7.2 g of carbon. Use this information to determine whether the compound is an alkane or an alkene. You **must** show your working. [3]

$$A_r(\text{H}) = 1 \quad A_r(\text{C}) = 12$$

Conclusion

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Examiner
only

11

